6.3.1 Initial insights

Bookmark this page

Week 06 due Feb 21, 2025 11:05 IST

Video

put a

superscript on it to indicate where it goes came from. And since we're going to

extend this to vectors with more than two elements, we'll actually end up with more

of these epsilons as well, so we're going to put a superscript on here as

well. And notice that was that all these epsilon in magnitude are bounded by the machine

epsilon. Okay? Now, at this point you can distribute this error term so that we

5:19 / 5:19

2.0x

Video

Download video file

Transcripts

Download SubRip (.srt) file

Download Text (.txt) file

Reading assignment

1/1 point (graded)

Read Unit 6.3.1 of the notes. [LINK]

done

Submit

Correct (1/1 point)

Discussion

Hide Discussion

Topic: Week 06 / 06.3.1

Add a Post

Show all posts

by recent activity

There are no posts in this topic yet.

Homework 6.3.3.1

1/1 point (graded)

Consider

$$x = \begin{pmatrix} \chi_0 \\ \chi_1 \\ \chi_2 \end{pmatrix} \text{ and } x = \begin{pmatrix} \psi_0 \\ \psi_1 \\ \psi_2 \end{pmatrix}$$

and the computation

$$\kappa := x^T y$$

computed in the order indicated by

$$\kappa := (\chi_0 \psi_0 + \chi_1 \psi_1) + \chi_2 \psi_2.$$

Employ the SCM to derive a result similar to that given in Example 6.3.1.1 for the computed result $\check{\kappa}$.

Which of the following holds: (where $|\epsilon_*^{(0)}|, |\epsilon_*^{(1)}|, |\epsilon_+^{(1)}|, |\epsilon_*^{(2)}|, |\epsilon_+^{(2)}| \leq \epsilon_{ma}$

Mark all correct answers

- [x] $\check{\kappa} = [(\chi_0 \psi_0 + \chi_1 \psi_1) + \chi_2 \psi_2]$
- [x] $\check{\kappa} = [[\chi_0 \psi_0 + \chi_1 \psi_1] + [\chi_2 \psi_2]]$
- [x] $\check{\kappa} = \left[\left[\begin{pmatrix} \chi_0 \\ \chi_1 \end{pmatrix}^T \begin{pmatrix} \psi_0 \\ \psi_1 \end{pmatrix} \right] + [\chi_2 \psi_2] \right]$
- [x] $\check{\kappa} = \left(\begin{pmatrix} \chi_0 \\ \chi_1 \end{pmatrix}^T \begin{pmatrix} (1+\epsilon_*^{(0)})(1+\epsilon_+^{(1)}) & 0 \\ 0 & (1+\epsilon_*^{(1)})(1+\epsilon \end{pmatrix} + \chi_2 \psi_2 (1+\epsilon_*^{(2)}) \right) (1+\epsilon_+^{(2)})$
- [x] $\check{\kappa} = \begin{pmatrix} \chi_0 \\ \chi_1 \end{pmatrix}^T \begin{pmatrix} (1+\epsilon_*^{(0)})(1+\epsilon_+^{(1)}) & 0 \\ 0 & (1+\epsilon_*^{(1)})(1+\epsilon_+^{(1} \end{pmatrix} + \chi_2 \psi_2 (1+\epsilon_*^{(2)})(1+\epsilon_+^{(2)})$
- [x] $\check{\kappa} = \begin{pmatrix} \chi_0 \\ \chi_1 \\ \chi_2 \end{pmatrix}^T \begin{pmatrix} (1+\epsilon_*^{(0)})(1+\epsilon_+^{(1)})(1+\epsilon_+^{(2)}) & \\ 0 & (1+\epsilon_*^{(1)}) \\ 0 & \end{pmatrix}$

✓

Solution:

For a complete solution, see the notes .

Submit

‹ Previous | Next ›

edX

[About](#)
Affiliates
edX for Business
Open edX
Careers
News

Legal

Terms of Service & Honor Code
Privacy Policy
Accessibility Policy
Trademark Policy
Sitemap
Cookie Policy
Your Privacy Choices

Connect

Idea Hub
Contact Us
Help Center
Security
Media Kit

GET IT ON Google play
Download on the App Store